

# JEFFREY MURRAY JR

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## TECHNICAL SKILLS

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**Languages:** Clojure, Java, Python, Dart, C#, C++, JavaScript, SQL

**Tools:** Amazon Web Services, GitHub, Docker, Kubernetes, Helm

**Focus:** Networking, Privacy, Security, Internet of Things

## EDUCATION

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**Master of Science** Cybersecurity Engineering 2024  
University of Washington | GPA 3.83

**Bachelor of Science** Computer Science and Software Engineering 2020  
University of Washington | GPA 3.56

## PROFESSIONAL EXPERIENCE

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**Software Developer** *Jan 2022 - Present*  
*SAP Concur – Bellevue, WA*

- Develop and maintain core services within the high-volume Profile Identity Platform.
- Collaborate cross-functionality to drive and ship major product features for customers.
- Spearheaded implementation of a distributed load testing with Locust to enhance production reliability.
- Achieved < 2s p99 latency for SCIM Filtering API (RFC 7644) querying 400M documents at 2M requests per day.

**Research Assistant** *Jul 2020 – Sep 2020*  
*University of Washington – Bothell, WA*

- Privacy Analysis of Data Anonymization of Customer Proprietary Network Information.
- Investigation of a large body of literature in privacy preserving data anonymization techniques.
- Delivered two workshop presentations and a whitepaper for our telecommunications partner.

**Software Developer Internship** *Oct 2019 – Feb 2020*  
*Apollo Video Technology – Bothell, WA*

- Developed a Flutter mobile application from scratch to decode live video streams.
- Utilized WebView plugin to render a decoded live video stream in HTML via JS.
- Layered Architecture for user interface handled asynchronous network calls in Dart.
- Collaboration with three team members to design and QA test application.

## TECHNICAL PROJECTS

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**Toward Easier Development of Privacy-Preserving Mobile Crowdsensing Applications** *Jan 2024 – Dec 2024*  
*Fully Homomorphic Encryption*

- Engineered GhostPeerShare for privacy-preserving comparison of encrypted video similarity.
- Abstracted Microsoft SEAL library for Dart plugin to manage C objects by reference within the memory stack.
- Facilitated proximity-based data exchange via Bluetooth and Wi-Fi on Android.
- Led a team of three through the full application lifecycle: design, implementation, and QA testing.
- Delivered a solution that enables secure video sharing while preventing the disclosure of user content or identity.

**Smart Mirror with Facial Recognition** *Sep 2021 – Dec 2021*  
*Internet of Things, Machine Learning*

- Provides authenticated users a unique, configurable content view upon request.
- Extended Magic Mirror platform in Node.js as a user interface on Raspberry Pi 4.
- Integrated Google Assistant for voice control and configurable wake word with Porcupine.
- Use of CUDA powered facial recognition in Python on Jetson Nano.
- Created RESTful API on Google Cloud Services to facilitate event-driven communication.